

Notes: Di Maggio and Kermani (RFS, 2017)

Credit-Induced Boom and Bust

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1 Big picture

- **Question.** Did an outward shift in mortgage credit supply to riskier borrowers contribute causally to the housing boom, the subsequent bust, and real economic fluctuations in the 2000s?
- **Setting.** The paper exploits a regulatory change in 2004: the Office of the Comptroller of the Currency (OCC) preempted national banks from state antipredatory lending laws (APLs). This deregulation mattered only in states that had APLs in place.
- **Core challenge.** Credit growth, house prices, and local employment all move with local demand. A county with stronger house-price growth may borrow more simply because demand is high, not because credit supply shifted out.
- **Main identification idea.** Compare counties in APL and non-APL states before and after 2004, while also exploiting cross-county variation in the preexisting local presence of national banks. Counties in APL states with a larger 2003 OCC lending share were exposed more strongly to the preemption shock.
- **Main first-stage result.** Comparing counties in the top and bottom deciles of OCC presence in APL states, the preemption increased annual lending by about 11% to 15% during 2004–2006.
- **Main second-stage results.** A 10% increase in loan origination raises annual house-price growth by about 3.3 percentage points and increases employment in nontradable sectors by about 2.2%. Delinquencies fall during the boom, but the same counties suffer larger delinquency increases in the bust.
- **Mechanism.** The deregulation allowed national banks to lend more aggressively to financially constrained and riskier households. These borrowers pushed up housing demand and collateral values, generating a local general equilibrium amplification.
- **Heterogeneity.** The effects are strongest in counties with more subprime borrowers, in places where housing is less affordable, and in inelastic housing markets.
- **Aggregate takeaway.** Under simple aggregation assumptions, the credit-supply channel can explain about 17% to 30% of the boom-bust in house prices and at least 30% of the fluctuations in employment and delinquency rates.
- **Why this paper matters.** It provides a direct, regulation-based credit-supply shock and shows that mortgage credit expansion was not just correlated with the boom-bust cycle but helped generate it.

2 Data and measurement (key objects)

2.1 Observed county-level and agency-level panel

- **Main unit of observation.** County-year, with supplemental agency $\times$ county $\times$ year observations for the first-stage analysis.
- **Mortgage lending.** Home purchase mortgage originations from HMDA. The main outcome is the log amount of newly originated purchase loans aggregated to the county-year level.
- **House prices.** CoreLogic county-level house price indices.

- **Employment.** County Business Patterns, with nontradable sectors defined as in Mian and Sufi (2014), excluding construction and other real-estate-related activities.
- **Delinquencies.** Federal Reserve Bank of New York Consumer Credit Panel; delinquency is defined as loans with payments at least 90 days late.
- **Borrower risk and financing environment.** Equifax measures of the fraction of borrowers with FICO scores below 680, county-level debt-to-income information from HMDA for the loan-level exercises, and BlackBox Logic securitization data for some robustness exercises.
- **Local controls.** Census-based population and income measures, plus housing supply elasticity from Saiz (2010).

2.2 Treatment variables and timing

- **APL indicator.** $APL_{g,t} = 1$ if state g has an antipredatory lending law in force at time t .
- **Post indicator.** $Post_{2004} = 1$ after the OCC preemption takes effect in 2004.
- **OCC exposure.** $FractionOCC_{i,2003}$ is the share of county i 's home-purchase loans originated by OCC-regulated lenders in 2003, before the deregulation.
- **Agency-level treatment.** In the within-county first stage, $OCC_a = 1$ if regulatory agency a is the OCC.
- **Why 2003 matters.** The 2003 OCC share is predetermined relative to the 2004 policy shock and highly persistent over time, so it acts like an exposure measure rather than a post-treatment outcome.
- **Key sample windows.** The main panel runs through the boom years, 2000–2006. Cross-sectional boom-bust exercises focus on 2003–2005 for the boom and 2007–2009 or 2008–2010 for the bust.

2.3 What the data say before estimation

- **Wide cross-county variation.** The fraction of OCC lenders in 2003 averages about 31%, with support from roughly 5% to 88%.
- **Persistence.** The county OCC share in 2003 and 2005 is correlated above 0.9, both in APL and non-APL states.
- **Balanced source of variation.** Counties with higher OCC shares differ along some dimensions (population, elasticity, subprime share), but those differences do not vary systematically with APL status, and there are no differential pretrends in key outcomes.

2.4 Empirical applications

The paper studies five related outcomes:

- mortgage loan origination,
- house price growth,

- employment in nontradable sectors,
- delinquency rates,
- and direct evidence on riskier lending through high-cost and high-DTI mortgages.

It also runs:

- event-study tests for pretrends,
- difference-in-differences and placebo exercises,
- boom-bust cross sections,
- border-census-tract tests,
- heterogeneity tests by subprime exposure, affordability, and housing elasticity,
- and back-of-the-envelope aggregate implications.

3 Models and empirical specifications (all equations + interpretation)

3.1 Regulatory first stage at the agency level

The paper first verifies that national banks themselves expand lending after the preemption in APL states:

$$\begin{aligned} \log(\text{Loan Amount})_{a,i,t} = & \lambda_{ia} + \eta_t + \beta_1(APL_{g,t} \times \text{Post}_{2004} \times \text{OCC}_a) \\ & + \beta_2(\text{OCC}_a \times \text{Post}_{2004}) + \beta_3(APL_{g,t} \times \text{OCC}_a) \\ & + \beta_4(APL_{g,t} \times \text{Post}_{2004}) + \beta_5\text{OCC}_a + \beta_6APL_{g,t} + \varepsilon_{a,i,t}. \end{aligned} \quad (1)$$

- **Interpretation.** This is a within-county, across-agency test. It asks whether OCC-regulated lenders expanded their lending differentially in APL states after 2004.
- **Why it matters.** It shows that the regulatory shock actually moved the supply of credit at the lender level before the paper studies downstream outcomes.

3.2 Main triple-difference county-level specification

The county-level lending equation is

$$\begin{aligned} \log(\text{Loan Amount})_{i,t} = & \lambda_i + \eta_t + \beta_0APL_{g,t} + \beta_1(APL_{g,t} \times \text{Post}_{2004}) \\ & + \beta_2(\text{FractionOCC}_{i,2003} \times \text{Post}_{2004}) + \beta_3(APL_{g,t} \times \text{FractionOCC}_{i,2003}) \\ & + \beta_4(APL_{g,t} \times \text{Post}_{2004} \times \text{FractionOCC}_{i,2003}) + \gamma X_{i,t} + \varepsilon_{i,t}. \end{aligned} \quad (2)$$

- **Coefficient of interest.** β_4 .
- **Interpretation.** This is a triple-difference estimator. It compares:
 - before versus after 2004,

- APL versus non-APL states,
- and counties with high versus low OCC presence.
- **Controls.** $X_{i,t}$ includes local income and population changes, subprime share, housing supply elasticity, and the local share of HUD-regulated independent mortgage lenders.
- **Economic meaning.** If $\beta_4 > 0$, then the preemption relaxed lending constraints specifically where the policy should matter most: places with APLs and a greater national-bank footprint.

3.3 Event-study specification

To test for pretrends and trace the boom-bust dynamics, the paper estimates

$$Y_{i,t} = \lambda_i + \eta_t + \sum_{\tau} \beta_{1\tau} APL_{g,2004} \mathbf{1}(\tau = t) + \sum_{\tau} \beta_{2\tau} \text{FractionOCC}_{i,2003} \mathbf{1}(\tau = t) + \sum_{\tau} \beta_{3\tau} APL_{g,2004} \times \text{FractionOCC}_{i,2003} \times \mathbf{1}(\tau = t) + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where the coefficient for 2003 is normalized to zero.

- **Interpretation.** $Y_{i,t}$ is alternately loan growth, house-price growth, nontradable employment, or delinquency.
- **Purpose.** The pre-2004 coefficients test the parallel-trends assumption; the post-2004 coefficients reveal whether the effect spikes during the boom and reverses in the bust.

3.4 Reduced form for local outcomes

For house prices, employment, and delinquency, the paper runs reduced forms analogous to equation (2):

$$Y_{i,t} = \lambda_i + \eta_t + \beta_0 APL_{g,t} + \beta_1 (APL_{g,t} \times \text{Post}_{2004}) + \beta_2 (\text{FractionOCC}_{i,2003} \times \text{Post}_{2004}) + \beta_3 (APL_{g,t} \times \text{FractionOCC}_{i,2003}) + \beta_4 (APL_{g,t} \times \text{Post}_{2004} \times \text{FractionOCC}_{i,2003}) + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (4)$$

where $Y_{i,t}$ is house-price growth, log employment in nontradables, or the delinquency rate.

- **Interpretation.** The same policy shock is used to see whether a credit supply expansion spills over into local asset prices, labor demand, and household distress.
- **Important economic point.** These are local general equilibrium effects, not just direct borrower-level effects. Higher credit availability raises housing demand, which raises collateral values and propagates the boom.

3.5 Instrumental-variables specification

The paper then interprets the triple interaction as an instrument for local credit supply:

$$Y_{i,t} = \beta \widehat{\log(\text{Loan Amount})}_{i,t} + \lambda_i + \eta_t + \gamma X_{i,t} + \varepsilon_{i,t},$$

where $\widehat{\log(\text{Loan Amount})}_{i,t}$ comes from the first stage in equation (2).

- **Interpretation.** The IV estimate turns the reduced form into an elasticity: how much do house prices, employment, or delinquency move when loan origination rises because of exogenous deregulation?
- **Why IV is needed.** Actual loan growth reflects both supply and demand. The regulatory interaction isolates the supply-driven component.

3.6 Difference-in-differences, placebo, and cross-sectional boom-bust designs

The paper supplements the main DDD with:

- a **difference-in-differences** design using only APL states, where the coefficient of interest is the post-2004 interaction with preexisting OCC presence;
- a **placebo test** on non-APL states, where the preemption should have no effect;
- **cross-sectional boom-bust regressions** using changes in outcomes from 2003 to 2005 and from 2007 to 2009 or 2008 to 2010;
- **loan-level risk regressions** for high-cost loans and high-DTI mortgages;
- and **border-tract regressions** comparing census tracts inside the same MSA but on opposite sides of state borders.
- **Interpretation.** These exercises are not separate identification strategies so much as stress tests for the main one.
- **What they check.** That the results are not driven by other lender types, general state shocks, unusual counties, or omitted regional heterogeneity.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Problem.** Places with stronger local demand will have more borrowing, faster house-price growth, and stronger nontradable employment even absent a credit supply shift.
- **Why that matters.** A simple regression of house prices or employment on local loan growth confounds credit supply with credit demand.
- **Further problem.** Counties with more national banks may look systematically different from counties with fewer national banks. States that adopted APLs may also differ from those that did not.

4.2 What identifies the model in the paper

- **Regulatory asymmetry.** The 2004 OCC preemption only changes national banks' constraints, and only in states that had APLs.
- **Exposure heterogeneity.** Counties differ in their preexisting exposure to national banks, measured by the 2003 OCC lending share.

- **Triple-difference logic.** The identifying variation is the extra post-2004 change in outcomes in high-OCC counties inside APL states, netting out:
 - post-2004 changes in high-OCC counties in non-APL states,
 - and post-2004 changes in low-OCC counties inside APL states.
- **Agency-level confirmation.** The within-county first stage shows that OCC-regulated lenders themselves expanded credit after preemption, which is exactly the mechanism the design relies on.

4.3 Why the identifying assumption is plausible

- **Persistence of OCC exposure.** The 2003 OCC share is highly persistent, so it behaves like a pre-policy local characteristic.
- **No differential pretrends.** Event-study graphs show no pre-2004 divergence in lending, house prices, employment, or delinquencies across treated and control counties.
- **Balanced observable trends.** Table 3 shows that OCC presence is not systematically related to pre-2004 trends in the main outcomes once the APL interaction is considered.
- **Controls for alternative channels.** The paper controls for subprime borrower shares, income, population, housing supply elasticity, and the local importance of independent mortgage lenders.
- **Placebo test.** In non-APL states, the same post-2004 OCC interaction does not move the outcomes.
- **Geographic tightening.** Border-census-tract regressions within the same metropolitan area produce similar results, reducing concern about regional omitted variables.

4.4 Mechanism checks

- **Riskier borrowers.** Effects are much stronger in counties with more subprime borrowers.
- **Tighter constraints.** Effects are stronger where houses are less affordable and where housing supply is inelastic.
- **Loan-level evidence.** OCC lenders increased high-cost loans and high-DTI loans after deregulation in APL states, which is exactly what a relaxation of anti-predatory constraints should imply.

4.5 Relation to the literature

- The paper differs from work that uses *exposure* to a presumed credit shock—for example housing supply elasticity or subprime borrower shares—because it tries to identify the *credit supply shock itself*.
- It complements work by Mian and Sufi and related studies on household credit and local demand by providing a deregulation-based instrument for mortgage supply.
- It also speaks to the literature on banking deregulation and the Great Recession by showing that regulation affected not only bank behavior but also local real outcomes.

5 Tables and figures

5.1 Table 1, Figure 1, and Figure 2: summary statistics and the source of exposure

Table 1
Summary statistics

		(1)	(2)	(3)	(4)	(5)
		<i>N</i>	<i>Mean</i>	<i>SD</i>	<i>Min</i>	<i>Max</i>
<i>Static characteristics</i>	Fraction of OCC lenders in 2003	2,219	0.31	0.13	0.05	0.88
	Elasticity of housing supply	770	2.36	1.25	0.6	12.15
	Log median income in 2003	2,219	10.54	0.23	9.82	11.44
	Log population in 2003	2,219	10.66	1.12	9.04	15.84
	Fraction borrowers with FICO <680 in 2003	2,219	0.47	0.11	0.2	0.8
<i>Change from 2003 to 2005</i>	Log median income	2,220	0.06	0.04	-0.15	0.28
	Log population	2,219	0.03	0.04	-0.32	0.20
	Fraction of loans securitized	2,211	0.14	0.08	-0.68	0.76
	Log loan amounts	2,219	0.36	0.23	-0.77	1.82
	House prices	679	0.22	0.15	-0.28	1.09
<i>Change from 2008 to 2010</i>	Log employment in nontradable sector	760	0.04	0.11	-0.40	0.94
	Log median income	2,220	-0.01	0.05	-0.28	0.21
	Log population	2,220	0.01	0.02	-0.07	0.13
	Log loan amounts	2,220	-0.27	0.22	-1.27	1.07
	House prices	714	-0.08	0.10	-0.49	0.16
	Log employment in nontradable sector	765	-0.06	0.07	-0.33	0.32

The table reports descriptive statistics for the main variables employed in our analysis. Loan Amount is computed using HMDA data, and denotes the value of mortgages to purchase a home by mortgage lenders in the period 2000–2011. Data on Population and Income are from the Census. House prices are from CoreLogic and are aggregated at the county level. The Fraction of OCC lenders in 2003 is the share of loans originated by all the mortgage lenders regulated by The Office of the Comptroller of the Currency (OCC) as of 2003, and is computed using data from HMDA. Employment data comes from County Business Pattern, and nontradable sectors are defined according to Mian and Sufi (2014). Delinquency is defined as at least 90 days late payments and comes from Federal Reserve Bank of New York Consumer Credit Panel. Fraction of Securitized loans come from BlackBox Logic, which covers 90% of the securitization market.

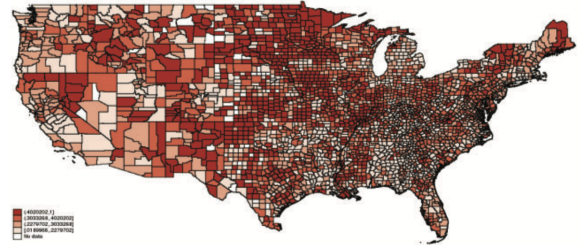


Figure 1
Fraction of lending done by national banks in 2003 in each county

Read:

- Table 1 shows that the preexisting county exposure to national banks varies enormously: the 2003 OCC share ranges from 5% to 88%, with a mean around 31%.
- Figure 1 visualizes this wide dispersion directly. This is the cross-county heterogeneity that powers the paper’s treatment-intensity design.
- Figure 2 shows that the OCC lending share is extremely persistent from 2003 to 2005, with correlation above 0.9. This supports treating the 2003 share as a predetermined exposure measure rather than a noisy post-treatment outcome.
- Table 1 also previews the boom-bust pattern: loan amounts, house prices, and nontradable employment rise on average in 2003–2005 and fall in the bust years.

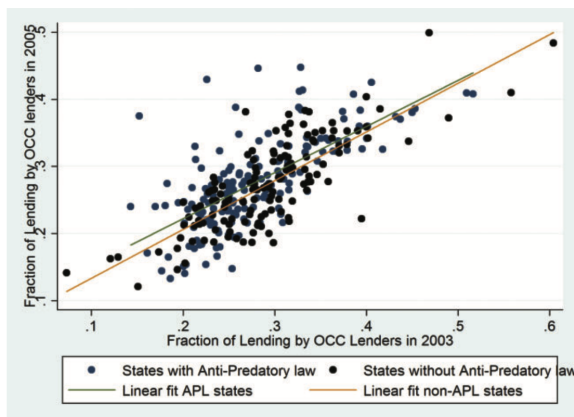


Figure 2
The relation between the fraction of lending done by national banks in 2003 (the year before the preemption rule) and in 2005 (the year after the preemption rule) for each county, differentiating between states with and without APLs

Table 2
Preemption of national banks and the amount of loans issued under each regulatory agency

	(1)	(2)	(3)	(4)
	<i>Log of loan amount</i>	<i>Log of loan amount</i>	<i>Log (Loan amounts/ Loan amounts in 2000)</i>	
APL _{g,t} × Post × OCC	0.0920*** (0.0246)	0.0899*** (0.0246)	0.11*** (0.03)	0.11*** (0.03)
APL _{g,t}	-0.00653 (0.0154)		-0.01	
APL _{g,t} × OCC	-0.0104 (0.0165)	-0.00962 (0.0165)	-0.00	-0.00 (0.02)
APL _{g,t} × Post	-0.0939*** (0.0188)		-0.10*** (0.02)	
OCC × Post	-0.0764*** (0.0143)	-0.0747*** (0.0142)	-0.09*** (0.02)	-0.09*** (0.02)
OCC			0.05*** (0.01)	0.05*** (0.01)
County-agency fixed effects	Yes	Yes		
Year fixed effects	Yes		Yes	
County fixed effects			Yes	
County-year fixed effects		Yes		Yes
Observations	120,218	120,034	89,170	89,170
R-squared	0.980	0.985	0.34	0.33

The table reports coefficient estimates of weighted least square regressions relating the amount of newly originated loans under each regulatory agency to the preemption of national banks with weights equal to the population of each county. Loan amounts is based on HMDA and is the amount of loans originated for purchasing a house aggregated for each regulatory agency at county level for each year. “APL_{g,t}” is equal to 1 if state *g* has an antipredatory lending law in place at time *t* and zero otherwise. “Post” is a dummy equal to one for years after 2004. “OCC” is equal to one if the regulating agency is OCC. The sample includes years from 2000 to 2006. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.2 Table 2: the OCC preemption moved national-bank lending

Read:

- Table 2 is the within-county, across-agency first stage. The triple interaction $APL \times Post \times OCC$ is about 0.09 to 0.11, meaning OCC-regulated lenders expanded lending by roughly 10% in APL states after 2004.
- This matters because the paper’s mechanism is lender-specific: the policy only loosened constraints for national banks, not for other lenders.
- The result survives county-agency fixed effects, county-year fixed effects, and growth specifications, so it is not merely picking up countywide demand conditions.

5.3 Table 3: why OCC exposure is usable

Read:

- Table 3 shows that counties with higher OCC shares differ in levels—they are somewhat less populated, more elastic in housing supply, and have fewer subprime borrowers.
- But the crucial interaction $APL \times \text{Fraction OCC}$ is generally insignificant in levels, and panel B shows no meaningful association with pre-2004 trends in the main outcomes.
- So the design is not relying on the assumption that high-OCC counties are identical to low-OCC counties; it only needs the extra treated-versus-control difference induced by the APL dimension to be clean.

Table 3
Presence of national banks as source of variation

A.	(1)	(2)	(3)	(4)		
	Median income in 2000	Population in 2000	Elasticity of housing supply	Fraction of subprime in 2000		
APL _{g,2004} × Fraction OCC	-0.20 (0.17)	-1.94 (1.31)	-2.04 (1.55)	0.12 (0.10)		
APL _{g,2004}	0.16*** (0.06)	1.10** (0.54)	0.50 (0.47)	-0.04 (0.03)		
Fraction OCC ₂₀₀₃	0.13 (0.09)	-3.21*** (0.72)	6.04*** (1.02)	-0.27*** (0.06)		
Constant	10.58*** (0.03)	13.15*** (0.29)	0.11 (0.31)	0.48*** (0.02)		
Observations	2,219	2,219	770	764		
R-squared	0.05	0.10	0.11	0.03		
B.	(1)	(2)	(3)	(4)	(5)	(6)
	Change in median income, 2001–2003	Change in population, 2001–2003	Change in loan amount, 2001–2003	Change in house prices, 2001–2003	Change in employment in nontradable, 2001–2003	Change in delinquency 2001–2003
APL _{g,2004} × Fraction OCC	0.02 (0.02)	-0.03** (0.02)	0.13 (0.11)	-0.08 (0.12)	-0.13 (0.08)	-0.26 (0.37)
APL _{g,2004}	-0.02*** (0.01)	0.02*** (0.01)	-0.04 (0.04)	0.06 (0.04)	0.04 (0.03)	0.07 (0.13)
Fraction OCC	-0.01 (0.01)	0.01 (0.01)	-0.38*** (0.07)	-0.26*** (0.07)	-0.09 (0.06)	0.54** (0.25)
Constant	0.03*** (0.00)	0.01** (0.00)	0.38*** (0.03)	0.21*** (0.02)	0.04** (0.02)	-0.20** (0.09)
Observations	3,075	2,219	3,078	892	747	2,219
R-squared	0.06	0.02	0.03	0.09	0.02	0.00

Panel A reports coefficient estimates of cross-sectional regressions relating the presence of national banks to several county characteristics. “Fraction OCC” is the fraction of purchase loans originated by OCC lenders in 2003. “ $APL_{g,2004}$ ” is equal to one if state g has an antipredatory lending law in place by 2004 and zero otherwise. “Elasticity” is a measure of elasticity of housing supply provided by [Saiz \(2010\)](#). “Fraction of Subprime” is the fraction of borrowers with FICO scores below 680 in 2000 for each county. Panel B reports coefficient estimates of cross-sectional regressions relating the presence of national banks to several trends in county characteristics. Change in Loan amount is based on HMDA and is the change in the amount of loans originated for purchasing a house aggregated at county level from 2001 to 2003. House prices are from CoreLogic. Employment data comes from County Business Pattern, and nontradable sectors are defined according to [Mian and Sufi \(2014\)](#). Delinquency is defined as at least 90 days late payments and comes from Federal Reserve Bank of New York Consumer Credit Panel. Robust standard errors, clustered at county level, are below the coefficients in parentheses. All regressions are weighted by the population of each county. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 4
Preemption of national banks and loan origination

	(1)	(2)	(3)	(4)	(5)
	Log of loan amount				
	Full sample		Counties with elasticity and FICO measure		
OCC	0.449*** (0.133)	0.472*** (0.120)	0.949*** (0.223)	0.915*** (0.189)	0.712*** (0.175)
$APL_{g,t} \times \text{Post}$	-0.174*** (0.0477)	-0.196*** (0.0416)	-0.324*** (0.0703)	-0.329*** (0.0589)	-0.290*** (0.0530)
$\text{Post} \times \text{Fraction OCC}$	-0.537*** (0.0987)	-0.446*** (0.0877)	-0.836*** (0.173)	-0.668*** (0.149)	-0.130 (0.126)
$APL_{g,t} \times \text{Fraction OCC}$	-0.242*** (0.0935)	-0.156 (0.101)	-0.258 (0.170)	-0.254 (0.158)	-0.323** (0.161)
$APL_{g,t}$	0.0546 (0.0342)	0.0384 (0.0363)	0.0676 (0.0544)	0.0841* (0.0508)	0.191*** (0.0692)
$\text{Log}(\text{median income})_{g,t}$		1.552*** (0.142)	1.725*** (0.157)	1.731*** (0.143)	1.484*** (0.156)
$\text{Log}(\text{population})_{g,t}$		1.191*** (0.156)	1.196*** (0.184)	1.180*** (0.159)	1.219*** (0.118)
Fraction of subprime $\times \text{Post}$				0.804*** (0.112)	0.950*** (0.118)
Elasticity $\times \text{Post}$					-0.0602*** (0.00917)
$APL_{g,t} \times \text{Fraction HUDg}$					-0.294** (0.133)
$\text{Post} \times \text{Fraction HUDg}$					0.357*** (0.1000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	21,564	15,533	5,348	5,348	5,348
R-squared	0.020	0.147	0.233	0.233	0.272
Number of counties	3,085	2,219	764	764	764

The table reports coefficient estimates of weighted least-squares regressions relating the amount of newly originated purchase loans to the preemption of national banks with weights equal to the population of each county. Loan amounts is based on HMDA and is the amount of loans originated for purchasing a house aggregated at county level for each year. “ $APL_{g,t}$ ” is equal to 1 if state g has an antipredatory lending law in place at time t and zero otherwise. “Post” is a dummy equal to one for years after 2004. “Fraction OCC” is the fraction of purchase loans originated by OCC lenders in 2003. “Elasticity” is a measure of elasticity of housing supply provided by [Saiz \(2010\)](#). “Fraction of Subprime” is the fraction of borrowers with FICO scores below 680 in 2000 for each county. “Fraction HUD” is the fraction of loans originated by HUD-regulated lenders in 2003 (a.k.a. independent mortgage lenders). The sample includes years from 2000 to 2006. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.4 Table 4 and Figure 3: main lending response and event-study dynamics

Read:

- Table 4 reports the county-level DDD estimates for loan origination. The coefficient on $APL \times Post \times \text{FractionOCC}$ is about 0.45 in the full sample and between 0.71 and 0.95 in the subsample with rich controls.
- The paper translates this into an economically intuitive comparison: counties in the top decile of OCC presence in APL states exhibit 11% to 15% higher annual lending after preemption than counties in the bottom decile.
- Figure 3 is the paper's most important credibility graph. There are no pretrends before 2004, but after the preemption the lending, house-price, and employment effects turn sharply positive, peak around 2005–2006, and later reverse.
- Delinquencies move in the opposite way at first: they fall through the boom and then rise in the bust, which is exactly the fragile-credit story the paper wants to tell.

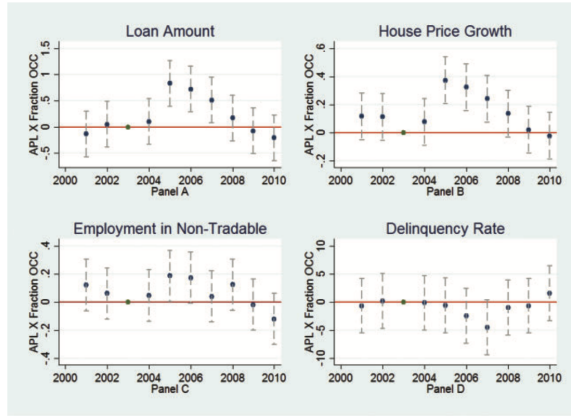


Figure 3
Yearly coefficients of the main dependent variables
This figure plots the coefficient between an indicator for the presence of APL, the fraction of loans originated by OCC lenders and a yearly dummy for the four main dependent variables. The coefficient for the year preceding the preemption rule, 2003, is normalized to zero.

Table 5
Preemption of national banks and house prices

	(1)	(2)	(3)	(4)	(5)	(6)
	House prices growth					
	Full sample		Counties with elasticity measure			
	IV					
APL _{g,t} × Post × Fraction OCC	0.247*** (0.0547)	0.215*** (0.0484)	0.273*** (0.0643)	0.273*** (0.0643)	0.231*** (0.0591)	
Instrumented log of loan amounts						0.330*** (0.110)
APL _{g,t} × Post	-0.107*** (0.0176)	-0.0993*** (0.0158)	-0.120*** (0.0197)	-0.120*** (0.0197)	-0.112*** (0.0181)	-0.0198* (0.0109)
Post × Fraction OCC	-0.200*** (0.0388)	-0.166*** (0.0335)	-0.174*** (0.0483)	-0.174*** (0.0483)	-0.0457 (0.0438)	0.0388 (0.0441)
APL _{g,t} × Fraction OCC	-0.117** (0.0597)	-0.0645 (0.0446)	-0.0640 (0.0574)	-0.0640 (0.0574)	-0.0858 (0.0611)	0.0168 (0.0567)
APL _{g,t}	0.0288 (0.0195)	0.0206 (0.0155)	0.0262 (0.0191)	0.0262 (0.0191)	0.0548* (0.0301)	-0.00302 (0.0302)
Fraction of subprime _g × Post			0.193*** (0.0370)	0.193*** (0.0370)	0.202*** (0.0349)	-0.196 (0.141)
Elasticity _g × Post					-0.0111*** (0.00314)	0.0153 (0.00980)
Log(median income) _{g,t}		0.200*** (0.0443)	0.239*** (0.0510)	0.239*** (0.0510)	0.200*** (0.0506)	-0.0781 (0.0918)
Log(Population) _{g,t}		0.219** (0.111)	0.162 (0.129)	0.162 (0.129)	0.170 (0.116)	-0.0656 (0.169)
APL _{g,t} × Fraction HUD _g					-0.0735 (0.0547)	-0.00573 (0.0529)
Post × Fraction HUD _g					0.120** (0.0507)	0.0442 (0.0443)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,244	5,322	3,258	3,258	3,258	3,258
R-squared	0.063	0.077	0.113	0.113	0.137	0.16
Number of counties	892	887	543	543	543	543

The table reports coefficient estimates of weighted least square regressions relating house prices to the preemption of national banks and the increase in the supply of loans induced by the preemption where the weights are given by the population of each county. House prices are from CoreLogic. “ $APL_{g,t}$ ” is equal to 1 if state g has an antipredatory lending law in place at time t and zero otherwise. “ Post ” is a dummy equal to one for years after 2004. “ Fraction OCC ” is the fraction of purchase loans originated by OCC lenders in 2003. “ Elasticity ” is a measure of elasticity of housing supply provided by Saiz (2010). “ $\text{Fraction of Subprime}$ ” is the fraction of borrowers with FICO scores below 680 in 2000 for each county. “ Fraction HUD ” is the fraction of loans originated by HUD-regulated lenders in 2003 (aka independent mortgage lenders). In column 6, “ $APL \times \text{Post} \times \text{Fraction OCC}$ ” is used as an instrument for the log of loan amounts. The sample includes years from 2000 to 2006. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.5 Table 5: effect of credit on house prices

Read:

- The reduced-form coefficient on $APL \times \text{Post} \times \text{FractionOCC}$ is positive and precisely estimated in every specification.
- In the IV column, a 10% increase in loan origination raises annual house-price growth by about 3.3 percentage points. Cumulated over 2004–2006, that corresponds to about a 10% rise in house

prices.

- The paper interprets this as a local general equilibrium effect: more credit to constrained borrowers raises housing demand and collateral values, which amplifies the price response.

5.6 Table 6 and Table 7: employment and delinquency

Table 6

Preemption of national banks and employment in nontradable sector

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Employment in nontradable sector</i>					
	<i>Full sample</i>		<i>Counties with elasticity measure</i>			
			<i>IV</i>			
APL _{g,t} × Post × Fraction OCC	0.207*** (0.0698)	0.161*** (0.0596)	0.179** (0.0707)	0.179** (0.0707)	0.146** (0.0665)	0.217** (0.0987)
Instrumented log of loan amounts						
APL _{g,t} × Post	-0.0696*** (0.0222)	-0.0607*** (0.0182)	-0.0715*** (0.0210)	-0.0715*** (0.0210)	-0.0682*** (0.0197)	-0.00730 (0.00995)
Post × Fraction OCC	-0.191*** (0.0560)	-0.103** (0.0448)	-0.0859 (0.0538)	-0.0859 (0.0538)	0.0227 (0.0519)	0.0473 (0.0431)
APL _{g,t} × Fraction OCC	-0.171*** (0.0538)	-0.0902* (0.0482)	-0.0637 (0.0593)	-0.0637 (0.0593)	-0.0956 (0.0585)	-0.0242 (0.0594)
APL _{g,t}	0.0544*** (0.0172)	0.0343** (0.0160)	0.0286 (0.0193)	0.0286 (0.0193)	0.0667*** (0.0234)	0.0231 (0.0270)
Log(median income) _{g,t}		0.287*** (0.0443)	0.310*** (0.0488)	0.310*** (0.0488)	0.296*** (0.0467)	-0.0212 (0.149)
Log(population) _{g,t}		0.893*** (0.0741)	0.954*** (0.0814)	0.954*** (0.0814)	0.925*** (0.0842)	0.647*** (0.160)
Fraction of subprime _g × Post			0.101*** (0.0352)	0.101*** (0.0352)	0.0945** (0.0378)	-0.127 (0.110)
Elasticity _g × Post					-0.00585** (0.00296)	0.00717 (0.00687)
APL _{g,t} × Fraction HUDg					-0.0921** (0.0411)	-0.0253 (0.0485)
Post × Fraction HUDg					0.145*** (0.0341)	0.0730 (0.0462)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,362	5,362	3,693	3,693	3,693	3,693
R-squared	0.014	0.226	0.287	0.287	0.298	0.158
Number of counties	790	790	537	537	537	537

The table reports coefficient estimates of WLS regressions relating employment in nontradable sectors to the preemption of national banks and the increase in the supply of loans induced by the preemption, with weights equal to the population of each county. Employment data comes from County Business Pattern, and nontradable sectors are defined according to Mian and Sufi (2014). "APL_{g,t}" is equal to one if state *g* has an anti-predatory-lending law in place at time *t* and zero otherwise. "Post" is a dummy equal to one for years after 2004. "Fraction OCC" is the fraction of purchase loans originated by OCC lenders in 2003. "Fraction of Subprime" is the fraction of borrowers with FICO scores below 680 in 2000 for each county. "Elasticity" is a measure of elasticity of housing supply provided by Saiz (2010). "Fraction HUD" is the fraction of loans originated by HUD-regulated lenders in 2003 (aka independent mortgage lenders). In column 6, "APL × Post × Fraction OCC" is used as an instrument for the log of loan amounts. The sample includes years from 2000 to 2006. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 7

Preemption of national banks and mortgages delinquencies

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Delinquency rates</i>					
	<i>Full sample</i>		<i>Counties with elasticity measure</i>			
			<i>IV</i>			
APL _{g,t} × Post × Fraction OCC	-0.874*** (0.328)	-0.869*** (0.309)	-1.830*** (0.442)	-1.830*** (0.442)	-1.545*** (0.431)	
Instrumented log of loan amounts						-2.170*** (0.583)
APL _{g,t} × Post	0.402*** (0.107)	0.420*** (0.101)	0.715*** (0.139)	0.715*** (0.139)	0.642*** (0.138)	0.0121 (0.0589)
Post × Fraction OCC	0.877*** (0.190)	0.683*** (0.178)	1.010*** (0.280)	1.010*** (0.280)	0.504* (0.268)	0.221 (0.266)
APL _{g,t} × Fraction OCC	0.0596 (0.282)	-0.00465 (0.284)	0.347 (0.391)	0.347 (0.391)	0.610 (0.397)	-0.0908 (0.317)
APL _{g,t}	-0.0520 (0.0943)	-0.0435 (0.0953)	-0.181 (0.128)	-0.181 (0.128)	-0.497*** (0.176)	-0.0820 (0.149)
Log(median income) _{g,t}		-1.930*** (0.325)	-2.316*** (0.372)	-2.316*** (0.372)	-1.846*** (0.406)	1.374 (0.887)
Log(population) _{g,t}		-0.827* (0.435)	-0.865* (0.515)	-0.865* (0.515)	-1.218** (0.519)	1.427* (0.853)
Fraction of subprime _g × Post			-0.843*** (0.304)	-0.843*** (0.304)	-1.292*** (0.312)	0.770 (0.624)
Elasticity _g × Post					0.0909*** (0.0220)	-0.0397 (0.0424)
APL _{g,t} × Fraction HUDg					0.804** (0.324)	0.167 (0.294)
Post × Fraction HUDg					0.169 (0.218)	0.943*** (0.290)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,533	15,533	5,348	5,348	5,348	5,348
R-squared	0.007	0.022	0.074	0.074	0.094	0.061
Number of counties	2,219	2,219	764	764	764	764

The table reports coefficient estimates of weighted least-squares regressions relating the percentage of delinquent mortgages to the preemption of national banks with weights equal to the population of each county. Delinquency is defined as at least 90 days late payments and comes from Federal Reserve Bank of New York Consumer Credit Panel. "APL_{g,t}" is equal to one if state *g* has an anti-predatory lending law in place at time *t* and zero otherwise. "Post" is a dummy equal to one for years after 2004. "Fraction OCC" is the fraction of purchase loans originated by OCC lenders in 2003. "Fraction of Subprime" is the fraction of borrowers with FICO scores below 680 in 2000 for each county. "Elasticity" is a measure of elasticity of housing supply provided by Saiz (2010). "Fraction HUD" is the fraction of loans originated by HUD-regulated lenders in 2003 (aka independent mortgage lenders). In column 6, "APL × Post × Fraction OCC" is used as an instrument for the log of loan amounts. The sample includes years from 2000 to 2006. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Read:

- Table 6 shows that the credit shock spills over to the real economy. The IV estimate implies that a 10% increase in loan issuance raises employment in nontradable sectors by about 2.2%.
- Table 7 shows that delinquencies actually decline in the boom when credit expands. The IV estimate implies roughly a 0.4 percentage-point reduction in delinquency rates during 2004–2006, about a 30% decline relative to the average boom-period delinquency rate.
- Taken together, these two tables say that easier credit boosted local demand and temporarily kept fragile borrowers afloat.

5.7 Table 8: difference-in-differences and placebo

Read:

- Panel A redoes the analysis using only APL states, comparing counties with different OCC exposure. The results for lending, house prices, employment, and delinquency remain strong.
- Panel B is the placebo. In non-APL states, the post-2004 OCC interaction does not generate significant effects in the outcomes that matter.

- This is a strong design check: the paper's results come from where the policy should have bite and disappear where it should not.

Table 8
Difference-in-differences and placebo test

A. Difference-in-differences in states with antipredatory lending laws

	(1)	(2)	(3)	(4)
	<i>Log of loan amount</i>	<i>House prices growth</i>	<i>Employment in nontradable sector</i>	<i>Delinquency rates</i>
Post × Fraction OCC	0.581*** (0.137)	0.132** (0.0610)	0.165*** (0.0611)	−0.696* (0.405)
APL _{g,t} × Fraction OCC	−0.297* (0.166)	−0.148* (0.0800)	−0.108* (0.0634)	0.793* (0.414)
APL _{g,t}	0.145** (0.0686)	0.0536 (0.0356)	0.0649** (0.0259)	−0.471** (0.184)
Post	−0.498*** (0.0798)	−0.149*** (0.0300)	−0.0850*** (0.0289)	0.847*** (0.257)
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	2,359	1,820	1,719	2,359
R-squared	0.488	0.128	0.433	0.142
Number of counties	337	260	252	337

B. Difference-in-differences in states without antipredatory lending laws

	(1)	(2)	(3)	(4)
Post × Fraction OCC	−0.113 (0.127)	−0.0154 (0.0430)	0.00433 (0.0605)	0.433 (0.284)
Post	−0.259***	−0.0768**	−0.0793**	0.537**
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	2,555	1,743	1,681	2,555
R-squared	0.411	0.221	0.384	0.104
Number of counties	365	249	243	365

The table reports coefficient estimates of weighted least square regressions relating the amount of newly originated purchase loans, house prices, employment in nontradable sectors, and delinquency rates to the preemption of national banks with weights equal to the population of each county. Panel A restricts attention only to the states that implemented an antipredatory law by January 2004. Panel B focuses on the states without an antipredatory law. Loan amounts is based on HMDA and is the amount of loans originated for purchasing a house aggregated at county level for each year. House prices are from CoreLogic. Employment data comes from County Business Pattern and non-tradable sectors are defined according to Mian and Sufi (2014). Delinquency is defined as at least 90 days late payments and comes from Federal Reserve Bank of New York Consumer Credit Panel. "APL_{g,t}" is equal to one if state *g* has an antipredatory lending law in place at time *t* and zero otherwise. "Post" is a dummy equal to one for years after 2004. "Fraction OCC" is the fraction of purchase loans originated by OCC lenders in 2003. Controls include: the fraction of borrowers with FICO scores below 680 in 2000 for each county; the measure of elasticity of housing supply provided by Saiz (2010); the log of median income, the log of population, the fraction of loans originated by HUD-regulated lenders (aka independent mortgage lenders) as of 2003 and their interaction with Post. The results are for years 2000 to 2006. All regressions are weighted using the total number of households in a county as weights. Robust standard errors, clustered at county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 9
Preemption of national banks and boom-bust cycle

A. Boom period

	(1)	(2)	(3)	(4)
	2003–2005			
	<i>Change in loan amount</i>	<i>Change in house prices</i>	<i>Change in employment in the nontradable sector</i>	<i>Change in delinquency rates</i>
APL _{g,2004} × Fraction OCC	0.736*** (0.198)	0.347* (0.189)	0.159** (0.0737)	−1.784*** (0.503)
APL _{g,2004}	−0.327*** (0.0634)	−0.167*** (0.0600)	−0.0584*** (0.0214)	0.693*** (0.159)
Fraction OCC	−0.233 (0.145)	−0.0938 (0.141)	−0.0281 (0.0600)	1.189*** (0.363)
Change in income	1.551*** (0.164)	1.354*** (0.160)	0.132** (0.0518)	−1.422*** (0.410)
Change in population	1.760*** (0.253)	0.649*** (0.205)	0.899*** (0.102)	−1.407** (0.585)
Fraction of subprime _g	0.640*** (0.105)	0.176** (0.0733)	0.0766** (0.0331)	−1.213*** (0.251)
Elasticity _g	−0.0455*** (0.00842)	−0.0542*** (0.00894)	−0.00333 (0.00286)	0.0310 (0.0194)
Fraction HUD _g	0.273*** (0.0976)	0.127* (0.0768)	0.0622** (0.0304)	0.131 (0.222)
Observations	764	543	528	764
R-squared	0.473	0.534	0.251	0.164

B. Bust period

	(1)	(2)	(3)	(4)	(5)
	2007–2009		2008–2010		
	<i>Change in loan amount</i>	<i>Change in loan amount</i>	<i>Change in house prices</i>	<i>Change in employment in nontradable sector</i>	<i>Change in delinquency rates</i>
APL _{g,2004} × Fraction OCC	−0.856*** (0.310)	−0.413 (0.284)	−0.500*** (0.166)	−0.215** (0.1000)	5.703* (3.312)
APL _{g,2004}	0.279*** (0.104)	0.177* (0.0933)	0.190*** (0.0543)	0.0660** (0.0317)	−2.401** (1.126)
Fraction OCC	0.963*** (0.229)	0.389** (0.191)	0.302** (0.146)	0.0201 (0.0567)	−6.836*** (2.464)
Change in income	1.187*** (0.313)	0.641** (0.267)	−0.153 (0.166)	0.120 (0.119)	1.018 (2.502)
Change in population	0.651 (0.523)	1.485*** (0.566)	1.102*** (0.337)	0.480** (0.224)	−24.07*** (5.830)
Fraction of subprime _g	−0.879*** (0.205)	−0.340*** (0.127)	−0.158** (0.0766)	0.0673 (0.0487)	4.667*** (1.559)
Elasticity _g	0.0698*** (0.0164)	−0.0254 (0.0158)	0.0375*** (0.00549)	0.00513 (0.00336)	−0.772*** (0.119)
Fraction HUD _g	0.163 (0.164)	−0.233** (0.101)	−0.258*** (0.0820)	−0.0813** (0.0367)	2.262* (1.220)
Observations	760	764	543	534	764
R-squared	0.281	0.148	0.323	0.058	0.348

The table reports coefficient estimates of weighted least square regressions relating the change in newly originated purchase loans to the preemption of national banks with weights equal to the population of each county. Panel A focuses on the period 2003–2005, while panel B focuses on the period 2007–2009 and 2008–2010. Loan amounts is based on HMDA and is the amount of loans originated for purchasing a house aggregated at county level for each year. "APL_{g,2004}" is equal to one if state *g* has an antipredatory lending law in place by 2004 and zero otherwise. "Post" is a dummy equal to one for years after 2004. "Fraction OCC" is the fraction of purchase loans originated by OCC lenders in 2003. "Elasticity" is a measure of elasticity of housing supply provided by Saiz (2010). "Fraction of Subprime" is the fraction of borrowers with FICO scores below 680 in 2000 for each county. "Fraction HUD" is the fraction of loans originated by HUD-regulated lenders in 2003 (aka independent mortgage lenders). Heteroskedasticity-robust standard errors are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.8 Table 9: boom-bust cycle in cross section

Read:

- Panel A shows that the counties hit harder by the preemption experienced a bigger boom in 2003–2005: more lending, faster house-price growth, stronger employment growth, and lower delinquency.
- Panel B shows that those same counties experience the sharper bust afterward. In particular, counties in the top decile of OCC presence in APL states had about 21% lower annual loan issuance in 2007–2009 than counties in the bottom decile.
- The counties that boomed the most on house prices and employment are the same counties that crash the most later, reinforcing the boom-bust interpretation.

Table 10
Antipredatory laws, national banks, and the amount of high cost loans

<i>A. High-cost loans</i>				
	(1)		(2)	
	<i>Log of loan amount</i>			
APL _{g,t} × OCC	0.391*** (0.0723)		0.386*** (0.0751)	
APL _{g,t}	-0.132*** (0.0316)			
Year fixed effects	Yes			
County fixed effects				
Agency fixed effects				
County-agency fixed effects	Yes		Yes	
County-year fixed effects			Yes	
Observations	59,398		57,589	
R-squared	0.942		0.950	
<i>B. High-DTI loans</i>				
	(1)	(2)	(3)	(4)
	<i>Average DTI</i>	<i>Average DTI</i>	<i>Log of loan amount</i>	<i>Log of loan amount</i>
APL _{g,t} × Post × OCC	0.0883*** (0.0169)	0.0884*** (0.0169)	0.0930*** (0.0290)	0.0880*** (0.0289)
APL _{g,t}	0.0584*** (0.0153)		0.0131 (0.0233)	
APL _{g,t} × OCC	-0.0393*** (0.0120)	-0.0393*** (0.0120)	-0.0252 (0.0193)	-0.0233 (0.0191)
APL _{g,t} × Post	-0.0865*** (0.0177)		-0.127*** (0.0249)	
Post × OCC	0.0898*** (0.00840)	0.0898*** (0.00841)	-0.0680*** (0.0189)	-0.0633*** (0.0187)
County-agency fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes		Yes	
County fixed effects				
County-year fixed effects		Yes		Yes
Observations	43,451	43,100	95,744	94,247
R-squared	0.780	0.911	0.975	0.982

Panel A reports coefficient estimates of weighted least square regressions relating the amount of newly originated high cost loans under OCC lenders and non-OCC lenders to the passage of antipredatory laws and the regulatory agency of loan originator. Loan amounts is based on HMDA and is the amount of high cost loans -loans with more than three percent spread rate - originated for purchasing a house aggregated for each regulatory agency at the county level for each year. Panel B reports coefficient estimates of weighted least-squares regressions relating the average debt-to-income ratio of newly originated loans (Columns 1 and 2) and the amount of newly originated loans with debt-to-income in the top tercile by OCC lenders and non-OCC lenders (Columns 3 and 4) to the passage of antipredatory laws and the regulatory agency of loan originator. Loan amounts is based on HMDA and is the amount of loans in the top tercile of debt-to-income ratio originated by each agency in each year. "APL_{g,t}" is equal to one if state *g* has an antipredatory lending law in place at time *t* and zero otherwise. "OCC" is equal to one if the regulating agency is OCC. The sample includes years from 2004 to 2007 (HMDA does not report high-cost loans before 2004). Robust standard errors, clustered at the county level, are below the coefficients in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.9 Table 10 and Table 11: mechanism and local geographic robustness

Read:

- Table 10 provides direct evidence on loan risk. After the preemption, OCC lenders in APL states increase high-cost loans by about 39%. They also increase average debt-to-income ratios by about 8.8% and the origination of top-tercile DTI mortgages by about 8% to 9%.
- This is important because it confirms that the policy shock did not simply increase lending mechanically; it relaxed constraints specifically in the risky segment of the mortgage market.
- Table 11 restricts the sample to census tracts near state borders and within the same MSA. The main interaction remains positive and significant for both loan amounts and house prices within 10- and 15-mile windows.
- So the results do not rely on broad geographic comparisons between very different places.

5.10 Table 12 and aggregate implications: who is hit hardest and how much it matters

Read:

- Table 12 shows that the effects are strongest in counties with a high share of subprime borrowers, in counties where housing is least affordable, and in inelastic housing markets.

- This matters because it lines up exactly with the paper’s mechanism: relaxing anti-predatory restrictions should matter most where borrowers are financially constrained and where additional credit translates most strongly into prices.
- In the paper’s back-of-the-envelope aggregation, the credit-supply channel explains about 17% to 30% of the boom-bust in house prices and at least 30% of the fluctuations in employment and delinquency.
- The authors are explicit that this aggregation is not a full structural general-equilibrium exercise, but it shows the mechanism is quantitatively meaningful, not just statistically significant.

6 Short summary

- **Conclusion.** The 2004 OCC preemption acted like an exogenous mortgage credit-supply shock in places where state anti-predatory lending laws had previously constrained national banks.
- **Identification insight.** The key variation is the interaction of APL status, the post-2004 period, and the county’s preexisting OCC lending share.
- **Empirical lesson.** More mortgage credit to riskier borrowers raised house prices and nontradable employment in the boom, but made the same local economies more fragile in the bust.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that deregulation-induced mortgage credit expansion had real effects on local housing markets and local employment, not just on loan quantities.
- It also shows that the same credit supply shock that lowered delinquencies in the boom raised them in the bust, highlighting an intertemporal fragility channel.
- More broadly, it demonstrates that banking regulation affected the macroeconomic path of the 2000s through the local supply of mortgage credit.

7.2 Economic intuition

- When national banks were exempted from state anti-predatory lending rules, they could extend more credit to marginal borrowers.
- Those borrowers increased housing demand, pushing up house prices. Rising house prices then relaxed collateral constraints and supported local spending, which lifted employment in nontradable activities.
- But the same expansion left households more leveraged and more exposed. When the boom reversed, the counties that had borrowed the most suffered the sharpest declines in lending, house prices, and employment, and the biggest increases in delinquency.

7.3 Takeaway

This paper is best read as a clean causal design linking mortgage-credit supply to the local boom-bust cycle of the 2000s. Its central message is that the housing boom was not just a demand story or a broad macro story: regulation shifted the supply of credit toward riskier borrowers, and that shift had first-order effects on house prices, local employment, and later financial distress. For research on housing, household finance, banking deregulation, and the Great Recession, this is a benchmark empirical paper.